

Lecture 2: Cardinality

Mathematical Analysis

1 Nested Sequence of Sets

Definition 1.1 (Nested Sequence of Sets). [def:nested-sequence-of-sets] A sequence of sets $(A_n)_{n \in \mathbb{N}}$ is *nested* if

$$\forall n \in \mathbb{N}, \quad A_{n+1} \subseteq A_n.$$

Theorem 1.1 (Nested Sequence Containment). [thm:nested-sequence-containment] Let $(A_n)_{n \in \mathbb{N}}$ be a nested sequence of sets. Then

$$\forall p, q \in \mathbb{N}, \quad p \leq q \implies A_q \subseteq A_p.$$

Proof. **TODO** □

2 Intervals

Definition 2.1 (Interval). [def:interval] A set $I \subseteq \mathbb{R}$ is an *interval* if, whenever $x, y \in I$ with $x \leq y$, every $z \in \mathbb{R}$ satisfying $x \leq z \leq y$ also belongs to I .

Remark 2.1 (Intervals as Sets). [rem:intervals-as-sets] In these notes, every interval is a set contained in $\mathbb{R}$; that is, every interval I satisfies $I \subseteq \mathbb{R}$.

Definition 2.2 (Open Interval). [def:open-interval] For $a, b \in \mathbb{R}$ with $a < b$, the *open interval* from a to b is

$$(a, b) = \{x \in \mathbb{R} : a < x < b\}.$$

Definition 2.3 (Closed Interval). [def:closed-interval] For $a, b \in \mathbb{R}$ with $a \leq b$, the *closed interval* from a to b is

$$[a, b] = \{x \in \mathbb{R} : a \leq x \leq b\}.$$

Lemma 2.1 (Endpoints of Contained Closed Intervals). [lem:contained-interval-endpoints] Let $I_1 = [a_1, b_1]$ and $I_2 = [a_2, b_2]$ be closed intervals. If $I_1 \subseteq I_2$, then

$$a_2 \leq a_1 \leq b_1 \leq b_2.$$

Proof. Since $a_1, b_1 \in I_1 \subseteq I_2$, definition 2.3 gives $a_2 \leq a_1$ and $b_1 \leq b_2$. Since $I_1 = [a_1, b_1]$ is a closed interval, definition 2.3 requires $a_1 \leq b_1$. Combining these inequalities gives the result. □

Lemma 2.2 (Endpoint Order of Nested Closed Intervals). [lem:nested-interval-endpoint-order] Let $I_k = [a_k, b_k]$ be a nested sequence of closed intervals. Then

$$n \leq m \implies a_n \leq a_m \leq b_m \leq b_n.$$

Proof. **TODO** □

Theorem 2.1 (Nested Intervals Property (NIP)). [thm:nip] Let $I_n = [a_n, b_n]$ be a sequence of closed and bounded intervals.

$$I_0 \supseteq I_1 \supseteq I_2 \supseteq \cdots \implies \bigcap_{n=0}^{\infty} I_n \neq \emptyset.$$

Proof. Let $A = \{a_n : n \in \mathbb{N}\}$. The nesting hypothesis gives

$$\forall n \in \mathbb{N}, \quad I_{n+1} \subseteq I_n.$$

For fixed $p \in \mathbb{N}$, define

$$P_p(r) \iff I_{p+r} \subseteq I_p.$$

$$P_p(0), \quad P_p(r) \implies I_{p+r+1} \subseteq I_{p+r} \subseteq I_p \implies P_p(r+1).$$

By Natural Number Induction and $p \leq q \iff \exists r \in \mathbb{N} : q = p + r$,

$$\forall p, q \in \mathbb{N}, \quad p \leq q \implies \exists r \in \mathbb{N} : \quad q = p + r \wedge I_q = I_{p+r} \subseteq I_p.$$

By comparability in an ordered set, $\forall n, m \in \mathbb{N}, (n \leq m) \vee (m \leq n)$. Hence lemma 2.1 gives

$$n \leq m \implies I_m \subseteq I_n \implies a_n \leq a_m \leq b_m,$$

$$m \leq n \implies I_n \subseteq I_m \implies a_n \leq b_n \leq b_m,$$

so

$$\forall n, m \in \mathbb{N}, \quad a_n \leq b_m.$$

Consequently,

$$a_0 \in A, \forall n \in \mathbb{N}, a_n \leq b_0 \implies A \neq \emptyset \wedge A \text{ is bounded above.}$$

By the Axiom of Completeness, choose $\alpha = \sup A$. The definition of supremum gives

$$\forall n, m \in \mathbb{N}, \quad a_n \leq \alpha \leq b_m.$$

Hence, by the definitions of a closed interval and indexed intersection,

$$(\forall n \in \mathbb{N}, \quad a_n \leq \alpha \leq b_n) \implies (\forall n \in \mathbb{N}, \quad \alpha \in I_n) \iff \alpha \in \bigcap_{n=0}^{\infty} I_n.$$

Therefore $\bigcap_{n=0}^{\infty} I_n \neq \emptyset$. □

Remark 2.2 (Intervals can be Infinitely Small). [rem:inf-small-intervals] If a size of the interval is infinitely small, ie $I = [a, b]$, $\forall \varepsilon > 0, b - a < \varepsilon$, this means that the interval is equivalent to a single point, ie $a = b$.

3 Decimal Expansion

Theorem 3.1 (Decimal Expansion). [thm:decimal-representation] Let $D = \{0, 1, \dots, 9\}$. Every digit sequence $a : \mathbb{N}_1 \rightarrow D$ determines a unique $x \in [0, 1]$:

$$\forall a : \mathbb{N}_1 \rightarrow D, \quad \exists! x \in [0, 1], \quad x = \sum_{k=1}^{\infty} a_k 10^{-k}.$$

Proof. We call the summation of the first n decimals s_n , ie

$$s_n = \sum_{k=1}^n a_k 10^{-k}$$

Since $\forall k \in \mathbb{N}_1, 0 \leq a_k \leq 9$, for all $n, m \in \mathbb{N}_1$ with $n \leq m$,

$$\begin{aligned} s_m &= s_n + \sum_{k=n+1}^m a_k 10^{-k} \\ &\leq s_n + 9 \sum_{k=n+1}^m 10^{-k} \\ &= s_n + 10^{-n} - 10^{-m} < s_n + 10^{-n}. \end{aligned}$$

In other words,

$$\forall n, m \in \mathbb{N}_1, \quad n \leq m \implies s_m \in [s_n, s_n + 10^{-n}]$$

By the same argument, we can also show that

$$\forall n, m \in \mathbb{N}_1, \quad n \leq m \implies s_m + 10^{-m} \in [s_n, s_n + 10^{-n}]$$

Where we can construct the following nested intervals

$$I_n = [s_n, s_n + 10^{-n}], \quad I_1 \supseteq I_2 \supseteq \cdots, \quad x = \sum_{n \in \mathbb{N}_1} a_n 10^{-n} = \bigcap_{n \in \mathbb{N}_1} I_n$$

According to the Nested Interval Theorem, after reindexing,

$$\bigcap_{n \in \mathbb{N}_1} I_n \neq \emptyset.$$

Also, notice that $I_n = [s_n, s_n + 10^{-n}]$, $s_n + 10^{-n} - s_n = 10^{-n}$. According to the Remark, we can conclude:

$$\forall \varepsilon > 0, \exists n \in \mathbb{N}_1 : I_n = [s_n, s_n + 10^{-n}] \text{ and } (s_n + 10^{-n}) - s_n = 10^{-n} < \varepsilon \implies \exists! x \in \bigcap_{n \in \mathbb{N}_1} I_n$$

Thus x is the unique number represented by the digit sequence a . □

Remark 3.1 (Decimal Expansion is not Unique). [rem:decimal-representation-not-unique] The decimal expansion of a real number is not unique. For example, $0.8999 \dots = 0.9000 \dots$

Theorem 3.2 (Uniqueness of Canonical Decimal Expansions). [thm:canonical-decimal-uniqueness] Let $D = \{0, 1, \dots, 9\}$. For every $x \in [0, 1)$, there exists a unique digit sequence $a : \mathbb{N}_1 \rightarrow D$ which is not eventually equal to 9 and satisfies

$$x = \sum_{n=1}^{\infty} a(n) 10^{-n}.$$

Here, “not eventually equal to 9” means

$$\neg \exists N \in \mathbb{N}_1 \forall n \geq N, \quad a(n) = 9.$$

Proof. **TODO** □

4 Archimedean Property

Theorem 4.1 (Archimedean Property). [thm:archimedean-property] $\mathbb{N}$ is not bounded above in $\mathbb{R}$, in other words, $\forall x \in \mathbb{R}, \exists n \in \mathbb{N}, n > x$

Proof. Assume for the sake of contradiction that $\mathbb{N}$ is bounded above in $\mathbb{R}$.

Since $0 \in \mathbb{N}$, we have $\mathbb{N} \neq \emptyset$. By the Axiom of Completeness, $\mathbb{N}$ has a supremum, denoted as $\alpha = \sup \mathbb{N}$.

By the definition of epsilon criteria, we have:

$$\forall x \in \mathbb{N}, x \leq \alpha, \quad \forall \varepsilon > 0, \exists x \in \mathbb{N}, \alpha - \varepsilon < x$$

Taking $\varepsilon = 1$, we obtain

$$\exists n \in \mathbb{N} : \quad \alpha - 1 < n \implies \alpha < n + 1.$$

Since $n + 1 = S(n) \in \mathbb{N}$, this contradicts that α is an upper bound of $\mathbb{N}$, namely

$$\forall m \in \mathbb{N}, \quad m \leq \alpha.$$

Therefore, $\mathbb{N}$ is not bounded above in $\mathbb{R}$. □

Theorem 4.2 (Archimedean Property (Reciprocal Form)). [thm:archimedean-reciprocal-form]

$$\forall y > 0, \exists n \in \mathbb{N}, 1/n < y$$

Proof. Take $x = 1/y$. Since $y > 0$, we have $x > 0$.

According to definition of Natural Numbers, $\exists n \in \mathbb{N}, n \geq x$, then $1/n \leq 1/x = y$. □

Definition 4.1 (Dense Subset). [def:dense-subset] Let $D \subseteq \mathbb{R}$. We say that D is *dense in* $\mathbb{R}$ if

$$\forall a, b \in \mathbb{R}, \quad a < b \implies \exists d \in D : \quad a < d < b.$$

Theorem 4.3 (Density of the Rational Numbers). [thm:rational-dense-in-reals] $\mathbb{Q}$ is dense in $\mathbb{R}$.

Proof. **TODO** □

Theorem 4.4 (Density of the Irrational Numbers). [thm:irrational-dense-in-reals] $\mathbb{R} \setminus \mathbb{Q}$ (ie, $\mathbb{I}$) is dense in $\mathbb{R}$.

Proof. **TODO** □

Remark 4.1. $\mathbb{Q}$ and $\mathbb{I}$ are both dense in $\mathbb{R}$. It does not mean that $\mathbb{Q}$ and $\mathbb{I}$ are having the same size. The size of infinite sets are defined in

5 Cardinality

Definition 5.1 (Cardinality). [def:cardinality] The *cardinality* of a set A , denoted by $|A|$, is the number of elements in A .

Definition 5.2 (Equinumerous Sets). [def:same-cardinality] Sets A and B are *equinumerous*, written $A \sim B$, if and only if there exists a bijective mapping $f : A \rightarrow B$:

$$A \sim B \iff \exists f : A \rightarrow B \text{ such that } f \text{ is bijective.}$$

Theorem 5.1. If $|A|, |B| \in \mathbb{N}$, ie, A and B are finite sets, then $|A| = |B| \iff A \sim B$

Definition 5.3 (Equivalence Relation). [def:equivalence-relation] A relation $\sim$ on a set A is called an *equivalence relation* if it satisfies the following properties:

1. Reflexive: $\forall a \in A, a \sim a$
2. Symmetric: $\forall a, b \in A, a \sim b \implies b \sim a$
3. Transitive: $\forall a, b, c \in A, a \sim b, b \sim c \implies a \sim c$

Theorem 5.2 (Cardinality is an Equivalence Relation). [thm:equivalence-relation-cardinality] *The Cardinality relation $\sim$ on the set of all sets is an equivalence relation.*

Example 5.1. $\mathbb{N} \sim \mathbb{N}_1$. Witness: $f : \mathbb{N} \rightarrow \mathbb{N}_1, f(n) = n + 1$

Remark 5.1. For infinite sets, the cardinality maybe still the same even if we remove some (or a finite number of) elements from the set.

Theorem 5.3 (Integers and Natural Numbers are Equinumerous). [thm:integers-natural-equinum] $\mathbb{Z} \sim \mathbb{N}$. *Witness:*

$$f = \begin{cases} n/2 & \text{if } n \text{ is even} \\ -(n+1)/2 & \text{if } n \text{ is odd} \end{cases}$$

Theorem 5.4 (Countability of the Rationals). [thm:rational-natural-equinum] $\mathbb{Q} \sim \mathbb{N}$. *Arrange the positive rationals in the Calkin–Wilf tree, where each a/b has children $a/(a+b)$ and $(a+b)/b$, and read it level by level:*

$$\frac{1}{1}; \quad \frac{1}{2}, \frac{2}{1}; \quad \frac{1}{3}, \frac{3}{2}, \frac{2}{3}, \frac{3}{1}; \quad \dots$$

Thus the levels contain $1, 2, 4, \dots$ elements. Let $r_1, r_2, \dots$ be this enumeration. Since every positive rational occurs exactly once, a witness $f : \mathbb{N} \rightarrow \mathbb{Q}$ is

$$f(n) = \begin{cases} 0, & n = 0, \\ r_{(n+1)/2}, & n \text{ is odd}, \\ -r_{n/2}, & n \text{ is even and } n > 0. \end{cases}$$

Theorem 5.5 (Equinumerosity of Intervals). [thm:interval-equinum] *Every interval which is not a single point is equinumerous to another interval which is not a single point.*

Proof. Witness:

$$f : (a, b) \rightarrow (c, d), f(x) = \frac{d-c}{b-a}(x-a) + c$$

□

Theorem 5.6 (Intervals and the Real Line). [thm:interval-real-line-equinum] *Every interval which is not a single point is equinumerous to $\mathbb{R}$.*

Proof. For $(-\pi/2, \pi/2)$, the function $f(x) = \tan(x)$ is a bijection from $(-\pi/2, \pi/2)$ to $\mathbb{R}$.

Notice that for every interval (a, b) which is not a single point, we can find a bijection from (a, b) to $(-\pi/2, \pi/2)$ by Theorem 5.5. Since bijections are transitive, we can conclude that every interval (a, b) which is not a single point is equinumerous to $\mathbb{R}$. □

6 Countability

Definition 6.1 (Countable Infinite Set). [def:countable-infinite-set] A set A is called *countably infinite* if $\mathbb{N} \sim A$.

Definition 6.2 (Countable Set). [def:countable-set] A set A is called *countable* if it is either finite or countably infinite.

Theorem 6.1 ($\mathbb{Z}, \mathbb{Q}$ are Countable). [thm:integers-rationals-countable] $\mathbb{Z}$ and $\mathbb{Q}$ are countable since they are equinumerous to $\mathbb{N}$ according to Theorem 5.3 and Theorem 5.4.

Theorem 6.2 (Cantor's Theorem). [thm:cantor-theorem] The interval $(0, 1)$ is uncountable.

Proof. Since $\mathbb{N} \sim \mathbb{N}_1$, suppose there exists a bijection $f : \mathbb{N}_1 \rightarrow (0, 1)$. By Theorem 3.2, choose the unique decimal expansion of each $f(n)$ which is not eventually equal to 9. This defines a function $g : \mathbb{N}_1 \rightarrow \mathbb{N}_1 \rightarrow \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

Then, we can define another function of decimal expansion

$$h(n) = \Upsilon(g(n)(n)), \quad n \in \mathbb{N}_1 \quad (1)$$

$$\Upsilon = \{(0, 1), (1, 2), (2, 3), (3, 4), (4, 5), (5, 6), (6, 7), (7, 8), (8, 1), (9, 1)\} \quad (2)$$

Then, we can construct

$$x = \sum_{n=1}^{\infty} h(n)10^{-n}.$$

Since $h(n) \in \{1, 2, \dots, 8\}$, we have $x \in (0, 1)$, and this is its unique canonical decimal expansion by Theorem 3.2. For every $n \in \mathbb{N}_1$, $h(n) \neq g(n)(n)$, so $x \neq f(n)$. Hence x is not in the image of f , contradicting the assumption that f is a bijection. Therefore, $(0, 1)$ is uncountable. $\square$

Theorem 6.3 ($\mathbb{R}$ is Uncountable). [thm:reals-uncountable] $\mathbb{R}$ is uncountable.

Proof. Assume that $\mathbb{R}$ is countable. According to Theorem 5.6, we can conclude that $(0, 1) \sim \mathbb{R} \sim \mathbb{N}$. According to Cantor's Theorem, $(0, 1)$ is uncountable, which contradicts the assumption that $\mathbb{R}$ is countable. Therefore, $\mathbb{R}$ is uncountable. $\square$

Theorem 6.4 (Countable Union of Countable Sets). [thm:countable-union-of-countable-sets] Given n countable sets $A_1, A_2, \dots, A_n$, the union $\bigcup_{i=1}^n A_i$ is countable.

Proof. We can prove this theorem by induction.

Let A and B be two countable sets. If A and B are finite, trivially it is countable. $\square$

7 Power Sets